



Mixed-Signal PCB Design

Analog meets digital — without killing each other

Partitioning • Grounding • ADC/DAC layout • Star points

Audience: Intermediate Engineering Students

ADC • DAC • Op-amps • Sensor signals • Star ground • PSRR



The Mixed-Signal Problem

Why analog and digital fight

Digital is noisy

Sharp edges inject
broadband noise into ground.

Analog is sensitive

μV -level signals
don't tolerate spikes.

Shared planes

Common GND lets noise
couple from digital $\rightarrow$ analog.

Sample rate matters

ADC clocks $\rightleftharpoons$ digital clocks
must be synchronised.

Reference voltage

Analog ref must be
MICROVOLT-clean.

Layout decides

Same schematic, two layouts
= 60 dB SNR difference.

Common Mixed-Signal Subsystems

Where students will meet this in practice

ADC + sensor

Temperature, pressure, current
flowing into MCU ADC.

Audio codec

Microphone in, speaker out.
Very clean rails required.

Motor control

ADC senses motor current.
Fast switching noise nearby.

DAC + output

Function generators,
precision references.

Battery gauge

m Ω current sense
digitised at 16 bit.

Instrumentation

Precision measurement
→ μ V resolution.

Board Partitioning

Physically separate analog from digital

- ▶ Draw a virtual line — "analog island" vs "digital land"
- ▶ All analog parts on one side: ADC, op-amps, references, sensors
- ▶ All digital on the other: MCU, memory, clocks, comms ICs
- ▶ Cross the boundary only at one well-defined point
- ▶ Bring digital ground & analog ground together at one star point
- ▶ Use a ferrite bead or $0\ \Omega$ to bridge ANALOG and DIGITAL grounds
- ▶ Keep clocks far from analog inputs
- ▶ Mark partitioning lines in silkscreen for the reviewer

Grounding Strategies

Single plane vs split — when to use which

Single solid GND (preferred)

- ▶ One continuous plane
- ▶ Route smartly — analog above analog, digital above digital
- ▶ Return current naturally finds shortest path
- ▶ Easier layout, simpler review
- ▶ Modern best practice (Henry Ott, Eric Bogatin)
- ▶ Works for 99% of mixed-signal designs

Split GND (only when needed)

- ▶ Two GND regions: AGND + DGND
- ▶ Single connection point (ferrite / 0 Ω)
- ▶ Used in very low-noise audio, instrumentation
- ▶ Easy to mess up — return paths must not cross split
- ▶ Only acceptable if you know exactly why
- ▶ Modern ADCs explicitly say: "use single GND"

Star Grounding for Sensitive Sections

Single point of return for ultra-clean signal paths

- ▶ Every analog return wire converges at ONE point — the 'star' or 'mecca'
- ▶ Star point is typically at the analog GND pin of the ADC or op-amp
- ▶ Prevents shared-impedance coupling: 'A's return current modulating B's voltage'
- ▶ Star is not always physical — sometimes implemented via a star-shaped via cluster
- ▶ Used at the highest-resolution measurements (24-bit ADCs, instrumentation amps)
- ▶ Don't apply to digital — return paths there should flow naturally

ADC Layout — Datasheet Layout Section

Always follow the vendor's recommended layout

- ▶ Place ADC over an UNBROKEN ground plane
- ▶ Analog input traces: short, equal length, equal width
- ▶ VREF pin: dedicated bypass cap, no other current path
- ▶ AVDD pin: separate LDO if possible
- ▶ AGND and DGND pins: connect at the IC, route plane underneath
- ▶ Differential inputs route as a tightly-coupled pair
- ▶ Anti-aliasing filter right before the input pins

Vendor app notes

TI / Analog Devices ADCs ship with a reference layout diagram. Copy it — vendors paid engineers months to optimise it.

VREF is sacred

VREF ripple becomes signal noise 1:1. Use a dedicated voltage reference (e.g. ADR4525) with a 10 μ F + 100 nF bypass right at the pin.

Powering Mixed-Signal ICs

Multiple rails, properly sequenced and decoupled

- ▶ Separate LDO for analog rail (AVDD) — never share with digital
- ▶ Pick high-PSRR LDO: LP5907, LT3045, ADP7104
- ▶ Ferrite bead between digital and analog rails as a fallback
- ▶ Decouple analog rail with 10 μ F (bulk) + 100 nF (HF) at every analog IC
- ▶ Sequence: AVDD UP before DVDD, AVDD DOWN after DVDD
- ▶ Use power-on reset (POR) to keep digital quiet during analog settling
- ▶ Watch DC accuracy: VREF spec is parts-per-million — track temperature drift

Sensor Signal Conditioning

From transducer to ADC input

Buffer / amplify

Op-amp matches sensor impedance to ADC.

Filter

RC or active LPF.
Anti-aliasing before ADC.

Level shift

Bipolar sensor → unipolar ADC.
Gain + offset.

Differential

Differential measurement rejects common-mode noise.

Isolation

Optocoupler / iCoupler for mains or noisy environment.

Cold-junction

Thermocouple → RTD compensation IC nearby.

Op-Amp Layout Rules

Stability and noise come from layout

- ▶ Feedback network IMMEDIATELY at the IC pins — short, direct routing
- ▶ Compensation cap pads on top side, traces < 5 mm
- ▶ Decoupling caps within 1 mm of supply pins
- ▶ Avoid long traces on the inverting input — picks up noise
- ▶ High-impedance input pins: keep traces tight, use guard ring if necessary
- ▶ Photodiode / very-high-impedance: cover with copper guard at virtual GND
- ▶ Route output trace away from input traces (avoid positive feedback)

Anti-Aliasing Filter Design

What happens above $f_s/2$ if you don't filter

- ▶ Nyquist: signals above $f_s/2$ fold back into your band
- ▶ Once aliased, no DSP can recover the original signal
- ▶ Filter MUST be analog, BEFORE the ADC sampling
- ▶ Passive RC: simple, single pole — good for slow ADCs
- ▶ Active LPF: 2nd / 4th order, sharper cutoff
- ▶ Sigma-delta ADCs have built-in digital filtering after oversampling

Cutoff rule

Set f_{cut} so the alias band attenuation > ADC dynamic range.
For a 12-bit ADC: > 72 dB rejection above $f_s/2$.

Common topology

Sallen-Key 2nd order LPF is the workhorse:
2 caps, 2 resistors, 1 op-amp.
Design tool: TI Filter Designer.

Clock & Reference Voltage

Two often-overlooked sources of noise

Clock & reference best practices

ADC Clock

- ▶ Jitter directly degrades ADC SNR
- ▶ Use a low-jitter crystal oscillator, not MCU PLL output
- ▶ Clean clock: dedicated LDO + ground island
- ▶ If clock comes from MCU, buffer it through a fan-out IC
- ▶ Route as a controlled-impedance trace, near a reference plane
- ▶ Keep clock far from analog signals

Voltage Reference

- ▶ Use a separate IC reference (ADR4xxx, REF50xx)
- ▶ Bypass with 10 μ F + 100 nF at the IC pin
- ▶ Watch noise spec: e.g. 3.3 μ Vpp 0.1-10 Hz
- ▶ Temperature coefficient (TC): ppm/ $^{\circ}$ C drives accuracy
- ▶ REF pin and ADC AVDD share an analog ground island
- ▶ Force/sense wiring on the REF pin if length matters

Common Mixed-Signal Mistakes

What review catches in 5 minutes

Digital signals over analog ground

Coupling noise into AGND area.

Routing clock over analog input

Capacitive injection — kills SNR.

Sharing AVDD with motor driver

Switching noise on the rail.

Single bypass cap on ADC

Need multi-value to cover frequency.

VREF trace as a normal signal

Pick up noise, lose accuracy.

Split planes with traces crossing

Return-path detour → degraded SNR.

Verification & Measurement

Tools to validate your mixed-signal design

Oscilloscope

Probe VREF, AVDD, input pin —
measure noise.

Spectrum analyzer

FFT of ADC capture —
find spurs from clocks.

Battery test

Run from battery (clean rail).
Is noise from PSU?

Two-tone test

Sweep input — measure THD,
SFDR, ENOB.

ADC eval-board

Reference layout proves
you're not the bottleneck.

Thermal chamber

Drift vs temperature
matters for precision.

Best Practices

Habits of every senior analog/mixed-signal engineer

- ▶ Partition analog and digital from the schematic stage
- ▶ Single solid GND plane (or split with clear single connection)
- ▶ Star point at the ADC GND pin
- ▶ Dedicated LDO + ferrite for analog rails
- ▶ Anti-aliasing filter on every ADC input
- ▶ Multi-value decoupling caps near every analog IC
- ▶ Voltage reference: dedicated IC + bypass
- ▶ Clock: low-jitter oscillator, dedicated supply
- ▶ Keep digital signals away from analog inputs
- ▶ Vendor reference layout — copy it
- ▶ Test point on AVDD, VREF, ADC input for debugging
- ▶ Document analog-digital boundary in silkscreen

Key Takeaways

Key takeaways from this lecture

Partition first

Layout decides analog performance.

One GND plane usually

Modern best practice.

Star at the ADC GND pin

Returns converge cleanly.

Dedicated analog rails

LDO + ferrite + decoup.

Filter before sampling

Anti-aliasing is mandatory.

Copy vendor layout

Reference designs are tested.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ Henry Ott — 'Electromagnetic Compatibility Engineering' (grounding bible)
- ▶ Eric Bogatin — 'Signal & Power Integrity Simplified' (mixed-signal chapters)
- ▶ Analog Devices — MT-031 'Grounding Data Converters and Solving the Mystery of "AGND" and "DGND"'
- ▶ TI app notes — SLOA060, SLAA559 — practical mixed-signal layout
- ▶ Linear Technology — Jim Williams Application Notes (gold standard)
- ▶ Robert Feranec — DDR + analog layout YouTube series
- ▶ Phil's Lab — practical analog frontend videos
- ▶ AD8616 / AD7124 datasheets — exemplary layout sections

Thank you • Build something this week.

Recommended Reference Designs

Boards worth ordering & dissecting

- ▶ Analog Devices EVAL-AD7124 — 24-bit ADC reference
- ▶ TI ADS131M08 evaluation module — multi-channel ADC
- ▶ Maxim MAX98357 audio codec eval — audio mixed-signal
- ▶ RPi 4 schematic — read its analog/digital partitioning
- ▶ Adafruit ADC breakouts (ADS1115) — entry-level mixed-signal
- ▶ AnalogDiscovery 2 / 3 — bench instrument with great mixed-signal example
- ▶ Robert Feranec's 'High-Speed Hardware Design Course' samples